

Satvik Singh

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Full-stack AI engineer with 3+ years building Go and TypeScript microservices, React front ends, and LLM and RAG features that run in production. Owner of Safe Security's cyber-risk scoring engine (72M calculations a month for 350+ enterprise tenants in 6 regions) and of its Temporal workflow platform, where production failures fell from 12.3% to 3.8% as run volume doubled.

SKILLS

- **Languages:** Go, TypeScript, JavaScript, Python, Java, SQL
- **AI and LLM:** RAG, hybrid vector and keyword search (AWS OpenSearch Serverless), embeddings, OpenAI GPT-4o/4.1, Claude on AWS Bedrock, prompt engineering, LLM guardrails, MCP tools, agentic workflow automation
- **Backend:** Microservices, distributed systems, event-driven design (RabbitMQ, Kafka), Temporal, REST APIs, Node.js, Express, Spring Boot, multi-tenancy, ABAC and RBAC, rate limiting and quotas
- **Data:** PostgreSQL, MySQL, DynamoDB, Redis, Cassandra, MongoDB, Snowflake, S3
- **Cloud and operations:** AWS (ECS Fargate, STS, KMS, SQS, MSK, OpenSearch), Docker, Terraform/OpenTofu, GitHub Actions, OpenTelemetry, Observe
- **Frontend:** React, TypeScript, design systems

EXPERIENCE

Safe Security, Software Development Engineer

Delhi, India · Jan 2025 to Present

Enterprise cyber-risk quantification and third-party risk management SaaS. Go and TypeScript microservices on AWS, React.

- Own the cyber-risk scoring engine behind every customer's risk posture: **72M risk-scenario calculations a month**, 6 production regions, 350+ enterprise tenants. Cut **p95 API latency 53% (604 to 281 ms)** and errors 58% with targeted database indexes, a memory-retention fix, keep-alive connections with bounded timeouts and a cache redesign.
- Own the Temporal-based workflow automation platform (Go) that runs customers' third-party-risk playbooks. Production failures fell from **12.3% to 3.8% while run volume nearly doubled**. Replaced an approved migration plan needing **1.5 hours of downtime with a drain-and-flip cutover** (one-second switch, no lost or duplicated runs) in four environments, and built its execution analytics (released Aug 2026).
- Now building per-tenant execution limits for the same platform, sized from 30 days of production telemetry: daily action quotas at run admission, Redis rate counters and concurrency leases that degrade safely when Redis is down, and caps on fan-out and workflow size. Core engine merged and in UAT. In parallel, moving the company's scheduled jobs onto Temporal Schedules.
- Designed and shipped the platform's **no-code HTTP integration node and encrypted credential store** (released Jul 2026), so customers can connect workflows to any external API themselves. Owned the security design across 6 services and the React UI: a DNS-pinned dialer that blocks DNS-rebinding SSRF, per-credential host allowlists, and KMS encryption so only ciphertext moves between services.
- Built and own the document pipeline behind AI questionnaire automation: secure upload and malware scanning, then chunking and embedding customer documents into a **RAG index on AWS OpenSearch Serverless** (hybrid vector and keyword search) used for LLM summarization, classification and auto-answering of security questionnaires, at **~25K scans and ~7.6K embedding jobs a month**. Added LLM input guardrails to workflow AI steps and an MCP tool that gives the company's AI assistant workflow analytics.
- Built a cross-tenant data exchange service from scratch (Go, single-table DynamoDB, cross-account AWS STS with confused-deputy protection, RabbitMQ with delayed retries and dead-letter queues): **~19,700 syncs a month across 4 regions, zero errors over 30 days**.
- Shipped attribute-based access control with organization-hierarchy data scoping, a P0 that unblocked Fortune 500 manufacturing and healthcare deals. Built a five-dimension executive vendor-risk view that aggregates nine services and degrades per dimension when one fails, plus its React drill-down UI: **2.2M API requests a month at p95 698 ms**.
- Led **13 production incidents** (2 major) from triage to root cause, including 575 failed customer runs in two hours traced one-to-one to malformed API calls, and a config type mismatch that had silently failed 2,000+ runs, fixed and verified four hours after escalation. Completed **700+ code reviews for 55 engineers**.
- Built developer tooling now used across engineering: a code knowledge graph pipeline live in **15 repositories** (daily signed-commit CI, per-service PRs routed by code ownership), a service dependency fact base engineers query before changing a service (unresolved dependencies cut from 79 to 11), and a Slack AI code reviewer teammates request by name.

Paisabazaar, Software Development Engineer

Gurugram, India · Jul 2023 to Jan 2025

India's largest consumer credit marketplace (loans and credit cards).

- Cut response time **~80%** on a core visit-ID service with a failsafe path, which also cut undelivered OTPs 10%. *Node.js, Redis, Kafka.*
- Built the company-wide authentication and authorization Java package for its microservices, with request monitoring. *Java, Redis.*
- Built user-consent workflows and core modules (gateway, webhooks, consent state) for the RBI Account Aggregator stack and onboarded a data provider onto it. *Spring Boot, Node.js.*
- Built a rate limiter that cut dropped requests 10% and internal voucher and refund tooling that sped up processing 20%. *Node.js, Redis.*

PROJECTS

- **InvestIQ** (personal, open source on GitHub): Flask and React investment app with an automated trading engine (6-layer risk checks, circuit breaker, kill switch), an XGBoost and random forest ensemble with walk-forward validation, FinBERT news sentiment and hierarchical risk parity portfolio optimization. *Python, scikit-learn.*

EDUCATION

National Institute of Technology, Meghalaya, B.Tech in Electrical and Electronics Engineering, CGPA 9.0/10

2019 to 2023